S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Practical No. 5

Aim: To implement the Random Forest algorithm and evaluate the model performance using appropriate metrics.

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Semester/Year : **6th / 3rd**

Academic Session : **2025-2026**

Date of Performance : _____

Date of Submission : _____

Aim: To implement the Random Forest algorithm and evaluate the model performance using appropriate metrics.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

- It's more accurate than the decision tree algorithm.
- It provides an effective way of handling missing data.
- It can produce a reasonable prediction without hyper-parameter tuning.
- It solves the issue of overfitting in decision trees.
- In every random forest tree, a subset of features is selected randomly at the node's splitting point.

THEORY:

A random forest is a supervised machine learning algorithm that is constructed from decision tree algorithms. This algorithm is applied in various industries such as banking and e-commerce to predict behavior and outcomes.This article provides an overview of the random forest algorithm and how it works. The article will present the algorithm's features and how it is employed in real-life applications. It also points out the advantages and disadvantages of this algorithm.

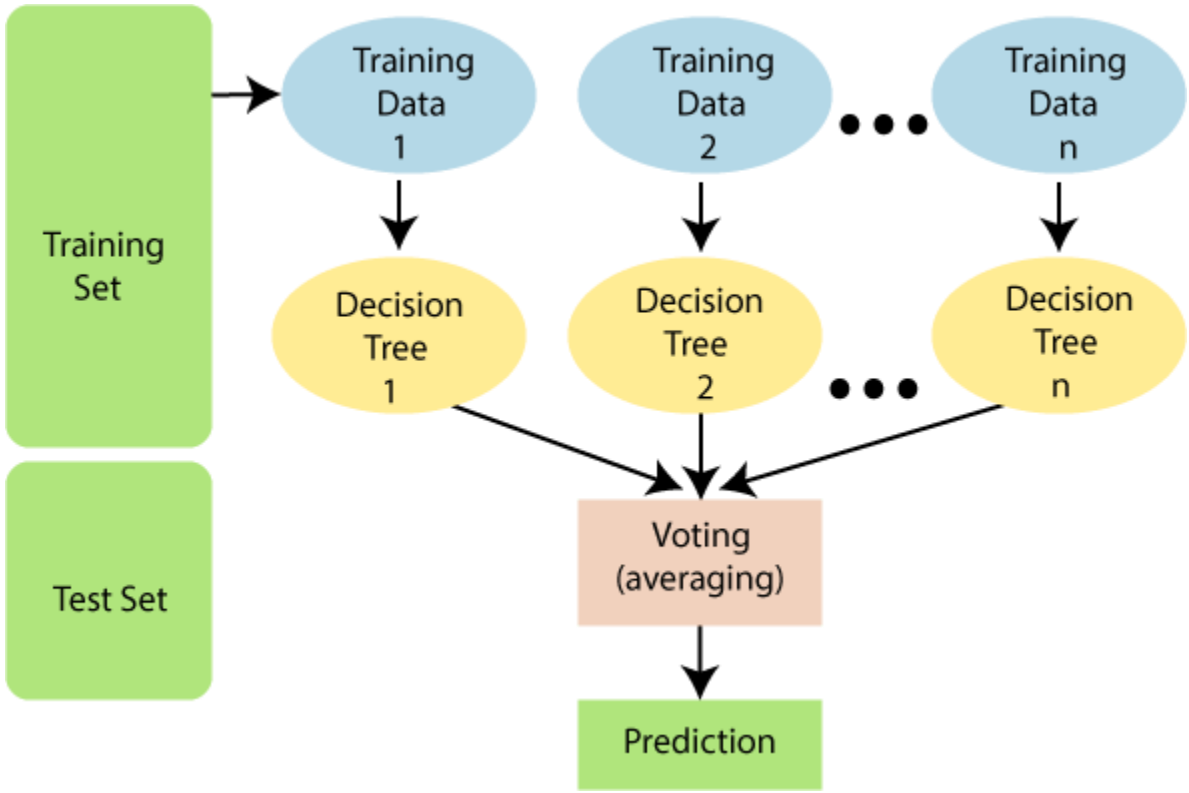
What is Random forest Algorithm?

A random forest is a machine learning technique that's used to solve regression and classification problems. It utilizes ensemble learning, which is a technique that combines many classifiers to provide solutions to complex problems. A random forest algorithm consists of many decision trees. The 'forest' generated by the random forest algorithm is trained through bagging or bootstrap aggregating. Bagging is an ensemble meta-algorithm that improves the accuracy of machine learning algorithms.

Why do we need a Random forest Algorithm?

Below are some points that explain why we should use the Random Forest algorithm:

- It takes less training time as compared to other algorithms.
- It predicts output with high accuracy, even for the large dataset it runs efficiently.
- It can also maintain accuracy when a large proportion of data is missing.



Algorithmic steps for Random Forest clustering

Random Forest works in two-phase first is to create the random forest by combining N decision tree, and second is to make predictions for each tree created in the first phase. The Random Forest works in two-phase first is to create the random forest by combining N decision tree, and second is to make predictions for each tree created in the first phase.

The Working process can be explained in the below steps and diagram:
Step-1: Select random K data points from the training set.
Step-2: Build the decision trees associated with the selected data points (Subsets).
Step-3: Choose the number N for decision trees that you want to build.
Step-4: Repeat Step 1 & 2.
Step-5: For new data points, find the predictions of each decision tree, and assign the new data points to the category that wins the majority votes.

PROGRAM CODE:

OUTPUT (SCREENSHOT):

CONCLUSION:

In this practical, we implemented the Random Forest algorithm and evaluated its performance using metrics such as accuracy, precision, recall, and F1-score.The model showed better accuracy and generalization compared to a single Decision Tree, as it reduces overfitting by combining multiple trees. Overall, Random Forest proved to be a robust and reliable classification method.

DISCUSSION AND VIVA VOCE:

1. When would you prefer Random Forest over a single Decision Tree in a production system?
2. Why is Random Forest less prone to overfitting compared to Decision Trees?
3. How do you choose the optimal number of trees (n_estimators) in an industry project?
4. What is the role of max_depth in Random Forest, and how does it impact model performance?
5. How does Random Forest handle high-dimensional data in real-world applications?
6. What are Out-of-Bag (OOB) samples, and how are they useful in industry?
7. How do you handle class imbalance when using Random Forest in fraud detection or medical diagnosis?
8. Why is feature importance from Random Forest useful in business decision-making?
9. What are the limitations of Random Forest in real-time systems?
10. Why might companies choose Gradient Boosting (XGBoost, LightGBM) over Random Forest?

REFERENCE

- https://www.analyticsvidhya.com/blog/2021/05/bagging-25-questions-to-test-your-skills-on-random-forest-algorithm/
- https://www.ibm.com/in-en/topics/random-forest#:~:text=Random%20forest%20is%20a%20commonly,both%20classification%20and%20regression%20problems.